Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

CODE :

```
# EXPERIMENT 1: FIND-S ALGORITHM

import pandas as pd

# Training dataset
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High',   'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High',   'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High',   'Strong', 'Cool', 'Change', 'Yes']
]

columns = [
    'Sky', 'AirTemp', 'Humidity', 'Wind',
    'Water', 'Forecast', 'EnjoySport'
]

df = pd.DataFrame(data, columns=columns)
print("Training Dataset:")
print(df.to_string(index=False))

# FIND-S algorithm
def find_s(dataset):
    attributes = dataset.iloc[:, :-1].values
    labels = dataset.iloc[:, -1].values

    # Start with the first positive example
    hypothesis = None

    for example, label in zip(attributes, labels):
        if label == 'Yes':
            if hypothesis is None:
                hypothesis = list(example)
```

```
            else:
                for i in range(len(hypothesis)):
                    if hypothesis[i] != example[i]:
                        hypothesis[i] = '?'

            print("Current hypothesis:", hypothesis)

    return hypothesis

final_hypothesis = find_s(df)

print("\nFinal Most Specific Hypothesis:")
print(final_hypothesis)
```

Outcome :

```
...  Training Dataset:
       Sky AirTemp Humidity   Wind Water Forecast EnjoySport
    Sunny    Warm   Normal Strong  Warm     Same        Yes
    Sunny    Warm     High Strong  Warm     Same        Yes
    Rainy    Cold     High Strong  Warm   Change         No
    Sunny    Warm     High Strong  Cool   Change        Yes
    Current hypothesis: ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same
    Current hypothesis: ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']
    Current hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']

    Final Most Specific Hypothesis:
    ['Sunny', 'Warm', '?', 'Strong', '?', '?']
```